

SpaceX

Missions Analysis Report

Analyst: Jennifer Onosatale

Date: 27th May 2026

Tools Used: Microsoft Excel, SQL, Tableau

Executive Summary

This report analyzes SpaceX mission data from 2006 to 2017 to uncover patterns in launch frequency growth, mission success rates, customer market distribution and rocket landing technology evolution — combining business analytics with physics domain knowledge.

Key finding reveals that Success rate improved from 0% in 2006 to consistently above 85% from 2014 onwards demonstrating remarkable reliability improvement. The United States dominates SpaceX's customer base generating 23 out of 41 missions — representing 56% of all launches. From 2014 onwards landing success improved dramatically from 40% in 2014 to 100% in 2017 — demonstrating SpaceX's rapid mastery of reusable rocket technology in just 3 years.

Based on these findings three strategic recommendations are provided – actively pursue international customers to reduce dependence on the US market, use their 85%+ mission success rate to attract new customers and expand into industries that need reliable launches, leverage their 100% landing success rate in 2017 to reduce costs through rocket reusability and pass on these savings to customer through lower pricing to attract more business.

Data Understanding

The dataset used in this analysis is the SpaceX Mission Dataset containing 41 rows and 16 columns spanning launch details from 2006 - 2017. It includes details on launch date, vehicle type, payload type, customer country, mission outcome, landing outcome.

	A	B	C	D	E	F	G
1	Flight Number	Launch Date	Launch Time	Launch Site	Vehicle Type	Payload Name	Payload Type
2	F1-1	24-Mar-06	22:30	Marshall Islands	Falcon 1	FalconSAT-2	Research Satellite
3	F1-2	21-Mar-07	01:10	Marshall Islands	Falcon 1	DemoSat	Classified
4	F1-3	03-Aug-08	03:34	Marshall Islands	Falcon 1	Trailblazer	Communication Satellite
5	F1-3	03-Aug-08	03:34	Marshall Islands	Falcon 1	PRESat, NanoSail-D	Research Satellites
6	F1-3	03-Aug-08	03:34	Marshall Islands	Falcon 1	Explorers	Human Remains
7	F1-4	28-Sep-08	23:15	Marshall Islands	Falcon 1	RatSat (DemoSat)	Classified
8	F1-5	14-Jul-09	03:35	Marshall Islands	Falcon 1	RazakSAT	Weather Satellite
9	F9-1	04-Jun-10	18:45	Cape Canaveral AFS LC-40	Falcon 9 (v1.0)	Dragon Spacecraft Qualification Unit	Classified
10	F9-2	08-Dec-10	15:43	Cape Canaveral AFS LC-40	Falcon 9 (v1.0)	SpaceX CRS (Dragon C1)	Space Station Supplies
11	F9-3	22-May-12	07:44	Cape Canaveral AFS LC-40	Falcon 9 (v1.0)	SpaceX CRS (Dragon C2+)	Space Station Supplies
12	F9-4	08-Oct-12	00:35	Cape Canaveral AFS LC-40	Falcon 9 (v1.0)	SpaceX CRS-1	Space Station Supplies
13	F9-4	08-Oct-12	00:35	Cape Canaveral AFS LC-40	Falcon 9 (v1.0)	Orbcomm-OG2	Communication Satellite
14	F9-5	01-Mar-13	15:10	Cape Canaveral AFS LC-40	Falcon 9 (v1.0)	SpaceX CRS-2	Space Station Supplies
15	F9-6	29-Sep-13	16:00	Vandenberg AFB SLC-4E	Falcon 9 (v1.1)	CASSIOPE	Communication/Research Satellite
16	F9-7	03-Dec-13	22:41	Cape Canaveral AFS LC-40	Falcon 9 (v1.1)	SES-8	Communication Satellite
17	F9-8	06-Dec-14	22:06	Cape Canaveral AFS LC-40	Falcon 9 (v1.1)	Thaicom 6	Communication Satellite

Figure 1: CLEAN_DATA Sheet showing dataset structure.

Data Cleaning

A thorough cleaning process was carried out before analysis was carried out on the dataset to ensure data integrity and consistency.

A duplicate check confirmed zero duplicate records in the dataset. A blank cell check across 16 columns using the COUNTBLANK formula returned eighty eight missing values. Six columns contained blank values; Payload Type(3), Payload Orbit(5), Customer Name(2), Customer Type(2), and Customer Country(2) blanks were filled with “Classified” as these likely represent secret or undisclosed missions. Payload Mass(8) blanks were filled with “Unknown” as average payload masses would be misleading. Failure Reason(33) blanks were filled with “N/A - Successful mission” as blank values in this column indicate the mission was successful and there was no failure report. Landing Type(13) blanks and Landing Outcome(20) blanks were filled “No Landing Attempted”. In Landing Type, seven cells contained “None” and were standardized to “ No Landing Attempted ” for consistency.

An abnormality check was carried out across all columns, no abnormalities were found. Three derived columns were created to support analysis - Year was extracted from Launch Date using =YEAR() Function for trend analysis, Mission Success and Landing Success were created using =IF() Function assigning 1 for success and 0 for failure and all other outcome to enable success rate calculation and landing success rate calculation.

Q	R	S
Year	Mission Success	Landing Success
2006	0	0
2007	0	0
2008	0	0
2008	0	0
2008	0	0
2008	1	0
2009	1	0
2010	1	0
2010	1	0

Figure 2: Derived Columns – Year, Mission Success, Landing Success.

Analysis and Findings

SpaceX Launch Frequency and Mission Success Rate

SpaceX failed all launches in 2006 and 2007 before achieving their first success in 2008. Launch frequency grew from 1 launch in 2006 to 10 launches in 2016 — a 900% increase. Success rate improved from 0% in 2006 to consistently above 85% from 2014 onwards demonstrating remarkable reliability improvement.

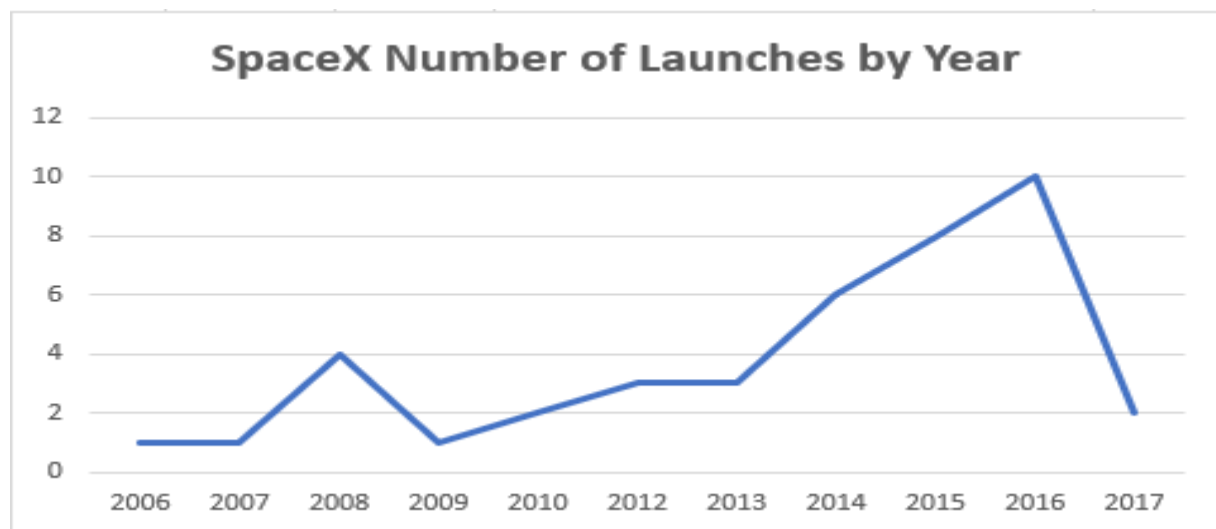


Figure 3: SpaceX Launches by Year



Figure 4: SpaceX Mission Success Rate

SpaceX Mission by Customer Country

The United States dominates SpaceX's customer base generating 23 out of 41 missions — representing 56% of all launches. This heavy reliance on US customers suggests SpaceX should diversify internationally to reduce business risk.

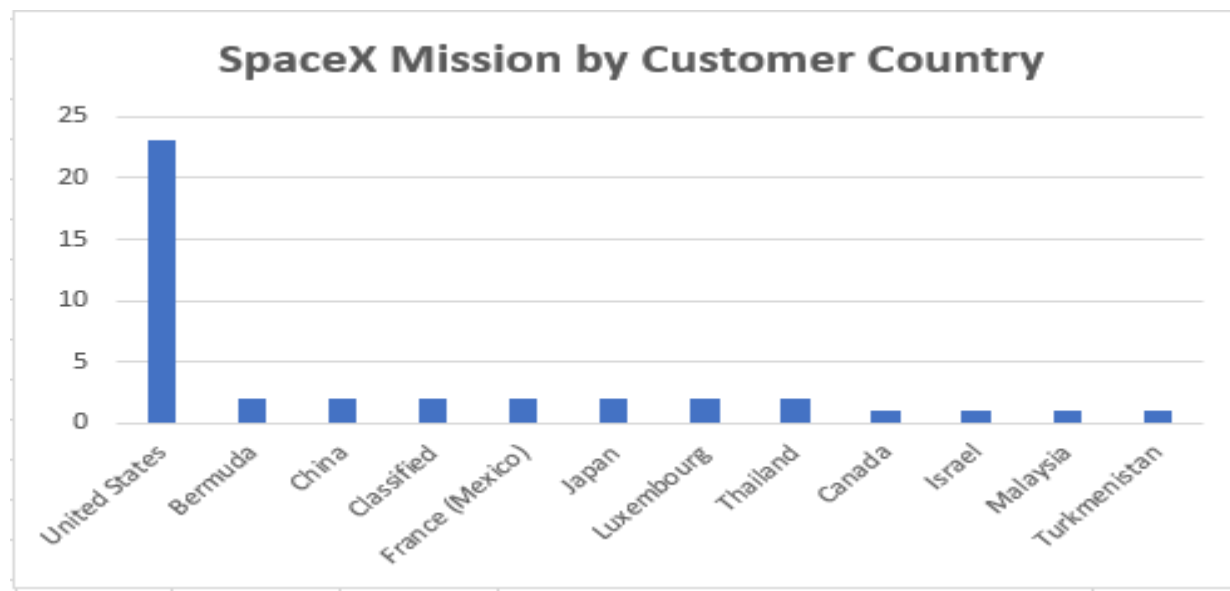


Figure 5: SpaceX Mission by Customer Country.

SpaceX Landing Success Rate by Year

SpaceX had a 0% landing success rate from 2006 to 2013 as reusable rocket technology didn't exist yet. From 2014 onwards landing success improved dramatically from 40% in 2014 to 100% in 2017 — demonstrating SpaceX's rapid mastery of reusable rocket technology in just 3 years.



Figure 6: SpaceX Landing Success Rate by Year.

SQL Findings

SQL analysis was conducted using SQLite online to validate and extend the Excel findings. Three queries were written and executed against the cleaned dataset.

Query 1 confirmed Pivot Table findings — SpaceX had 1 launch in 2006 with 0% success rate growing to 10 launches in 2016 with 90% success rate.

Query 2 confirmed Pivot Table findings — United States dominates SpaceX customer base with 23 out of 41 missions representing 56% of all launches.

Query 3 confirmed Pivot Table findings — From 2014 onwards landing success improved dramatically from 40% in 2014 to 100% in 2017 — demonstrating SpaceX's rapid mastery of reusable rocket technology in just 3 years.

```
1 SELECT
2 year,
3 COUNT(flight_number) AS Total_Launches,
4 ROUND(AVG(mission_success)*100,1) AS Success_Rate
5 FROM SpaceX_Clean
6 GROUP BY year
7 ORDER BY year ASC
8
```

year	Total_Launches	Success_Rate
2006	1	0
2007	1	0
2008	4	25
2009	1	100
2010	2	100

Figure 7 : SpaceX Launch Frequency and Mission Success Rate

```

9 SELECT
10 customer_country,
11 COUNT(flight_number) AS Total_Launches
12 FROM SpaceX_Clean
13 GROUP BY customer_country
14 ORDER BY Total_Launches DESC
15

```

customer_country	Total_Launches
United States	23
Thailand	2
Luxembourg	2
Japan	2
France (Mexico)	2
Classified	2

Figure 8: SpaceX Mission by Customer Country.

```

16 SELECT
17 year,
18 ROUND(AVG(landing_success)*100,1) AS Landing_Success_Rate
19 FROM SpaceX_Clean
20 GROUP BY year
21 ORDER BY year ASC
22

```

year	Landing_Success_Rate
2006	0
2007	0
2008	0
2009	0
2010	0
2012	0

Figure 9 : SpaceX Landing Success Rate by Year.

Tableau Dashboard

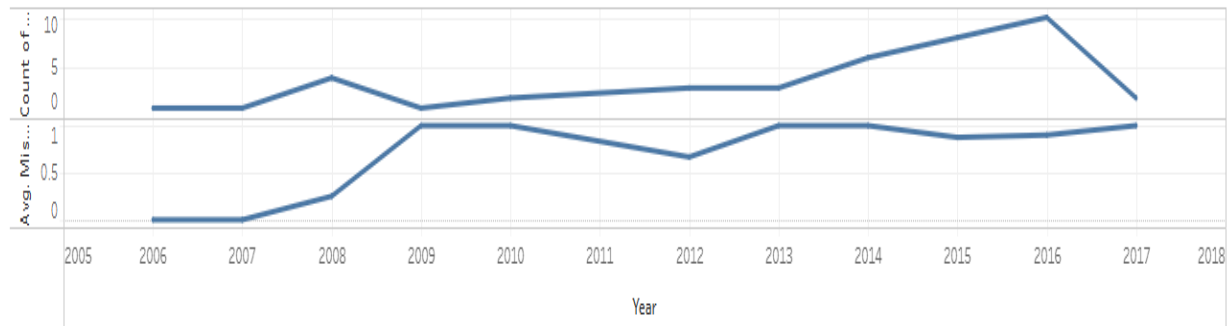
An interactive dashboard was created combining three visualizations to provide a comprehensive view of SpaceX Mission Analysis. The dashboard includes a line chart showing launch frequency and mission success rate, a bar chart showing mission by customer country, and a line chart showing landing success rate by year.

The dashboard is published publicly and accessible at:

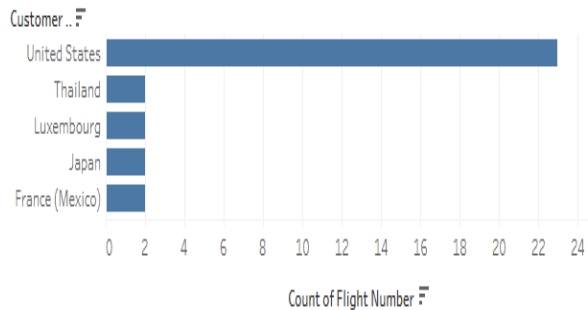
https://public.tableau.com/views/SpaceXMissionAnalysis/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

SpaceX Mission Analysis 2006 - 2017

SpaceX Launch Frequency and Mission Success Rate 2006 - 2017



SpaceX Missions by Customer Country



SpaceX Landing Success Rate by Year



Figure 10: SpaceX Mission interactive Tableau Dashboard

Recommendations

- SpaceX should actively pursue international customers to reduce dependence on the US market which currently represents 56% of all missions.
- SpaceX should use their 85%+ mission success rate to attract new customers and expand into industries that need reliable launches.

- SpaceX should leverage their 100% landing success rate in 2017 to reduce costs through rocket reusability and pass on these savings to customers through lower pricing to attract more business.

Ethics and Limitations

Dataset Coverage

This analysis is limited to SpaceX mission from 2006 to 2017 and does not represent SpaceX's complete mission history. Findings and recommendations may not reflect SpaceX's current operational capabilities and success rates which have likely improved significantly beyond 2017

Classified Data

Several missions contained classified payload and customer information recorded as 'Classified' or 'Unknown'. This missing information may introduce bias into the analysis as classified missions likely represent government or military contracts which could significantly affect customer distribution and payload mass findings.

Data Source

The dataset was sourced from Kaggle and may not be officially verified by SpaceX – findings should be validated against official SpaceX records before making business decisions.

Recommendation Impact

Recommendation to reduce pricing based on cost savings from reusable rockets should be carefully considered as aggressive pricing could impact SpaceX employees and suppliers who depend on current revenue level.

